(1.) Course Title: INFPHD618 Approximation theory (complex exam preparatory)	**Credits: 6**
Course **Classification: Mandatory chosen**	
Theoretical or Practical Nature of the Course, *'training character'*[13]: 100% theoretical (credit%)	
Type of Class: practice and **number of hours**: **2/week** in the given semester, *(If the course is not (only) taught in Hungarian, then **the language**: **English**)* **Additional** (specific**) methods and characteristics**[2] used in delivering the knowledge *(if any)*: -	
Assessment Method (Exam / Midterm Grade / **Other**[3]): practical course mark **Additional** (specific) **methods**[4] applied in knowledge assessment *(if any)*: -	
Course Placement in Curriculum (Semester Number): 4.	
Prerequisites (*if any*):	

Course Description: **A concise yet informative description of the knowledge** to be acquired.

1. Fundamental theorems of Approximation Theory: Weierstrass' First and Second Approximation Theorem. The reciprocal coherence between the two Weierstrass Theorems.
2. The basic problems of the approximation theory in normed linear spaces. Approximation in Banach and Hilbert spaces: existence, uniqueness and the determination of the best approximation. The minimum property of the Fourier projection in Hilbert spaces.
3. Function spaces. Function spaces of continuous functions. Function spaces of integrable functions. Weighted function spaces. Existence and uniqueness of the best approximation in these spaces.
4. Chebyshev approximation: Chebyshev alternation theorems. Deriving of Chebyshev polynomials and their fundamental properties.
5. Modulus of continuity and smoothness in he function spaces $C2\pi$ and $Lp2\pi$. Direct or Jackson type theorems on the approximation of periodic functions.
6. Inverse or Bernstein type theorems on the approximation of periodic functions. Approximation by algebraic polynomial. The order of the approximation by Bernstein polynomials. The problem of the saturation. The Ditzian-Totik modulus of smoothness. Direct and inverse theorems.
7. The approximation properties of the trigonometric Fourier partial sums. Summation processes of the trigonometric Fourier series. The Natanson-Zuk theorem for the uniform convergence. Classical uniformly convergent processes. The order of the convergence for the summations of trigonometric Fourier series.
8. Constructions of uniformly convergent algebraic polynomial sequences. Orthogonal polynomilas. Classical orthogonal polynomilas. Fourier series with respect to orthogonal polynomials and their summations.
9. Interpolation with algebraic and trigonometric polynomials. Lagrange, Hermite and Hermite-Fejér interpolation. The convergence of the interpolation processes. The role of the Lebesgue function. Weighted interpolation. Summation of discrete linear processes and their convergence.

2-5 Most Important Required and Recommended Literature
List of required and recommended literature (notes, textbooks) with bibliographic data (author, title, publication details, (possibly pages), ISBN).

1. E. W. Cheney: Introduction to Approximation Theory, McGraw-Hill, New York, (1966).
2. E. W. Cheney and W. Light: A Course in Approximation Theory, Brooks/Cole Publ. Comp., 2000.
3. R. A. DeVore: The Approximation of Continuous Functions by Positive Linear Operators, Lecture Notes in Mathematics, 293, Springer-Verlag, Berlin, 1972.
4. R. A. DeVore and G.G. Lorentz: Constructive Approximation, Springer-Verlag, New York, 1993.
5. G.G. Lorentz, M. von Golitschek and Y. Makovoz: Constructive Approximation: Advanced Problems, Springer-Verlag, New York, 1996.
6. I.P. Natanson, Constructive Function Theory, I. ,II., III., Frederick Ungar Publishing Co., New York, 1964.
7. P. P. Petrushev and V. A. Popov: Rational Approximation of Real Functions, Cambridge Univ. Press, New York, 1987.
8. A. F. Timan: Theory of Approximation of Functions of a Real Variable, Macmillman, New York, 1963. Reprint, Dover, New York.
9. V.K. Dzyadyk and I.A. Shevchuk, Theory of Uniform Approximation of Functions by Polynomials, Walter de Gruyter, Berlin - New York, 2008.

List of Prescribed Professional Competencies and Competency Elements

(knowledge, skills, etc., KKK Section 8) that the course significantly contributes to developing (highlight the most important ones).

a) Knowledge

- Has a research-level understanding of the general and specific characteristics, major directions, and precisely developed boundaries of the subject area within the discipline, as well as the agreed and debated interrelations.
- Creatively comprehends the interrelations, theories, and conceptual frameworks and terminologies of the given field of study.[1]
- Possesses research methodology knowledge necessary for independent research in the given discipline/field.

b) Skills

- Capable of creative analysis of the field, synthesizing comprehensive and specific interrelations with a new perspective, and conducting appropriate evaluative and critical activities.
- Able to apply and further develop the specialized learning and problem-solving methods of the field.
- Can creatively elaborate on the novel, previously unknown practical implications of theoretical issues.
- Capable of planning and executing new projects, conducting research in the field, and developing new techniques and approaches.

[1]**Section 108**, Article 37 of the National Curriculum Act (Nftv.) 37. lesson: a session (lecture, seminar, exercise, consultation) requiring the personal involvement of the instructor to fulfil the academic requirements specified in the curriculum, the duration of which is at least forty-five minutes and not more than sixty minutes.
[2] e.g. case studies, role play, thematic presentations, etc.
[3] e.g. continuous debriefing, mid-year report
[4] e.g. request for case studies, thematic papers, essays, business, organisational plans, etc.

- Able to identify unforeseen professional problems and uncover the detailed theoretical and practical background necessary for their resolution at a research level.
- Can construct and communicate new relations relevant to the field, as well as comprehensive interrelations significant for personal and communal existence.

c) Attitudes

- Represents and further develops the relations that contribute to the process of human self-creation based on the field's characteristics.
- Possesses an interest and learning ability that enable identifying and solving currently unpredictable research problems in the field.
- Demonstrates a firm professional commitment, a continuous dedication to exploring new approaches, and an acceptance of the necessity for persistent effort.

d) Autonomy and Responsibility

- Builds and initiates new knowledge areas with creative independence and proposes new practical solutions.
- Participates in formulating theoretical and practical questions with a leadership role and high-level cooperation.
- Engages as an equal discussion partner with professionals in the field.
- Takes responsibility for raising and addressing new ethical questions concerning theoretical and practical aspects of the profession.

Course responsible (name, position, academic degree): **Dr. László Szili, professor**

Instructor(s) involved in teaching the course (if any) (name, position, academic degree):
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